



BLOOD GROUPS & Rh FACTOR

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LEARNING OBJECTIVES

By the end of this lecture, students should be able to:

- Define what bloods groups are & explain their importance in blood transfusions.
- Identify the main ABO blood group system (A, B, AB, O).
- Describe the antigens & antibodies involved in blood typing.
- Explain why mismatched transfusions can be harmful.
- Define the concept of Rh factor & distinguish between Rh+ & Rh- blood types.

INTRODUCTION

When blood transfusions from one person to another were first attempted, immediate or delayed agglutination and hemolysis of red blood cells(RBCs) often occurred resulting in typical transfusion reactions that frequently led to death. But why it occurs in some cases & not in other cases remained a mystery until the discovery of blood groups by the Austrian scientist **Karl Landsteiner** in 1901. Karl Landsteiner was honored with Noble Prize in 1930 for this discovery.

Blood types of different people have different antigenic & immune properties so that antibodies in the plasma of one blood type will react with antigens on the surfaces of RBCs of another blood type.

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At least 30 commonly occurring antigens & hundreds of other rare antigens have been found on the surfaces of cell membranes of human blood cells each of which can at times cause antigen-antibody reactions. Two particular types of antigens also called **agglutinogens** (because of their capacity to cause agglutination of RBCs) are much more likely than others to cause blood transfusion reactions are named as **A antigen** & **B antigen**. Landsteiner also noticed the presence of corresponding antibodies or **agglutinins** in the plasma & named them as **A antibody** & **B antibody**. A particular agglutinogen & the corresponding agglutinin cannot be present together.

LANDSTEINER'S LAW

Landsteiner's law states that:

- If a particular agglutinogen(antigen) is present in the RBCs, corresponding agglutinin(antibody) must be absent in the serum.
- If a particular agglutinogen is absent in the RBCs, the corresponding agglutinin must be present in the serum.

ABO SYSTEM(O-A-B BLOOD TYPES)

The ABO blood group genetic locus has three alleles(A, B & O) means three different forms of the same gene. The O type of allele is functionless so it causes no significant type O agglutinin on the RBCs. While the type A & type B do cause strong agglutinogens on the cells. Thus, the O allele is recessive to both the A & B alleles.

As each person has only two sets of chromosomes, only one of these alleles is present on each of the two chromosomes in any individual. However, the presence of three different alleles means that there are six possible combinations of alleles:

- OO
- OA
- OB
- AA
- BB
- AB

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These combinations of alleles are called the *genotypes* & each person is one of the six genotypes.

BLOOD GROUPS

Based on the presence or absence of antigen A & antigen B, blood is divided into four groups:

1. A group
2. B group
3. AB group
4. O group

TABLE 18.1 Blood Groups and Donor Compatibility

Blood Group	Antigens on RBCs	Antibodies in Plasma	Donor's Group
A	A	anti-B	A, O
B	B	anti-A	B, O
AB	A, B	nil	AB, A, B, O
O	nil	anti-A, B	O

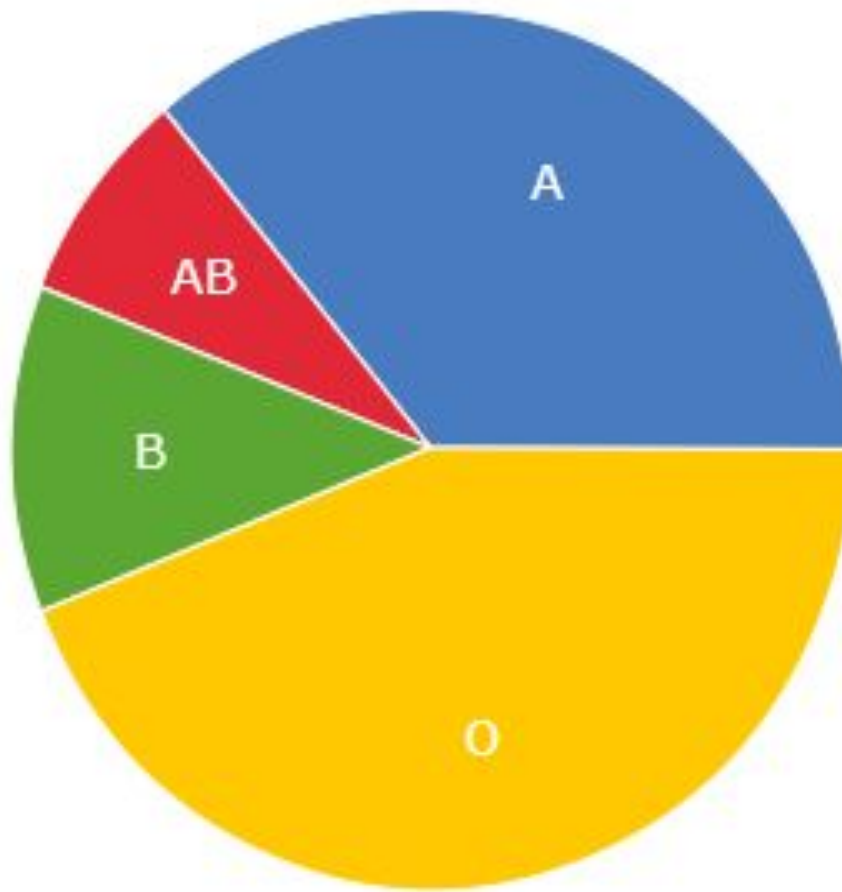
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A person with genotype OO produces no agglutinogen, therefore the blood type is O. A person with genotype OA or AA produces type A agglutinogens, therefore has blood type A.

Genotypes OB & BB produces B agglutinogen so give type B blood & genotype AB gives type AB blood.

ABO Blood Group		
Genotype	Phenotype(blood type)	
AA	A	
AO	A	
BB	B	
BO	B	
OO	O	
AB	AB	

RELATIVE FREQUENCIES OF DIFFERENT BLOOD GROUPS



Blood Type

■ A, 36%

■ AB, 8%

■ B, 12%

■ O, 44%

AGGLUTININS

Agglutinins are **gamma-globulins** as are other antibodies & are produced by the same cells that produce antibodies to any other antigens.

Most of the agglutinins are IgG & IgM immunoglobulin molecules.

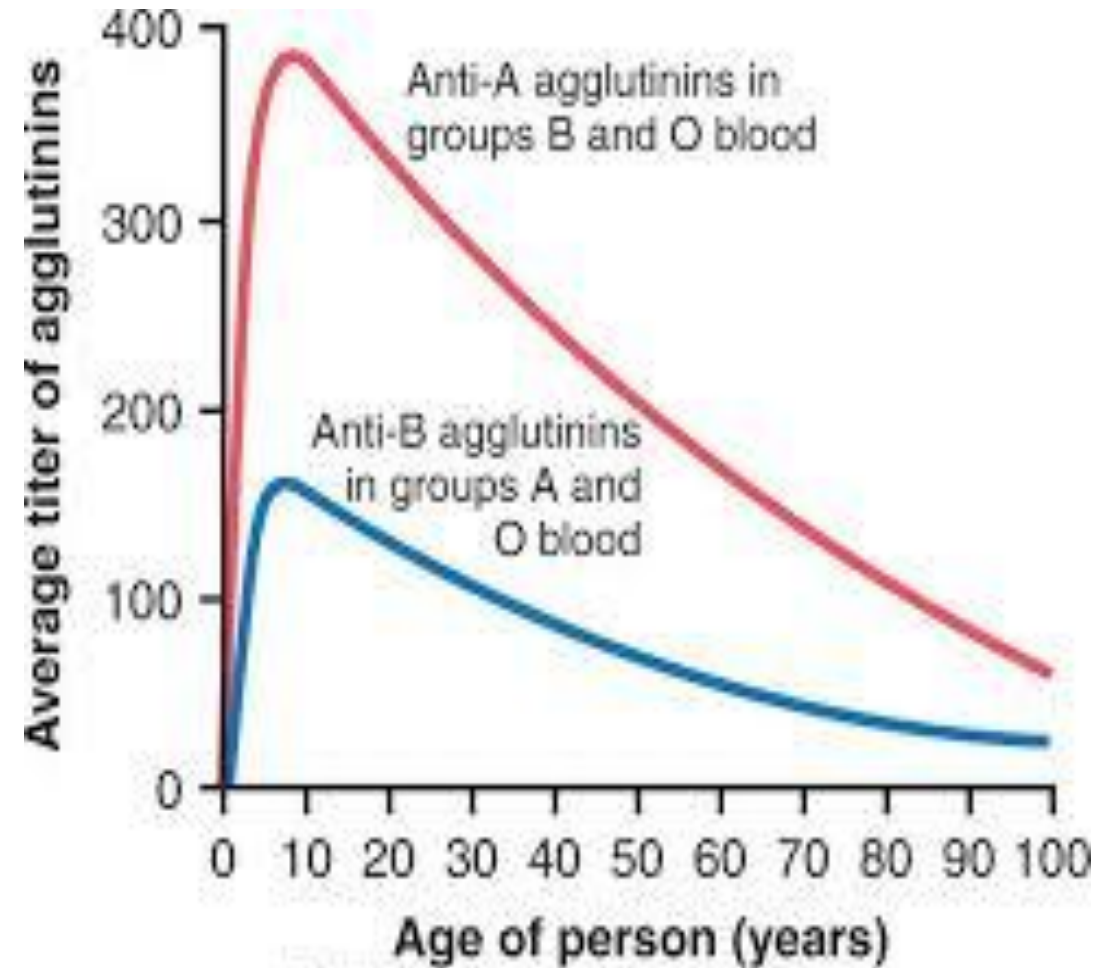
Why are these agglutinins produced in people who do not have the respective agglutinogens in their RBCs?

Possibly it is due to small amounts type A & B antigens that enter the body in food, bacteria & other ways & these substances initiate the development of anti-A & anti-B agglutinins.

TITER OF AGGLUTININS AT DIFFERENT AGES

Immediately after birth the quantity of agglutinins in the plasma is almost zero showing that agglutinins formation occurs almost entirely after birth.

At 2 to 8 months after birth, an infant begins to produce agglutinins (anti-A agglutinins when type A agglutinogens are not present in the cells & anti-B agglutinins when type B agglutinogens are not present in the RBCs). A maximum titer is usually reached at 8 to 10 years of age & this titer then gradually declines throughout the remaining years of life.



AGGLUTINATION

Agglutination is the collection of separate particles like RBCs into clumps or masses.

Agglutination occurs when an antigen is mixed with its corresponding antibody called *isoagglutinin* i.e when A antigen is mixed with anti-A or when B antigen is mixed with anti-B.

AGGLUTINATION IN TRANSFUSSION REACTIONS

When bloods are mismatched i.e anti-A or anti-B plasma agglutinins are mixed with RBCs that contain A or B agglutinogens respectively, RBCs agglutinate as a result of agglutinins attaching themselves to the RBCs. As the agglutinins have two binding sites(IgG) & ten binding sites(IgM), a single agglutinin can attach to two or more RBCs at the same time causing the cells to be attached together by agglutinins forming clumps. These clumps then plug small blood vessels throughout the circulatory system.

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During the ensuing hours to days, physical distortion of cells or attack by phagocytic WBCs destroy the membranes of agglutinated RBCs releasing hemoglobin into the plasma called **hemolysis** of RBCs.

ACUTE HEMOLYSIS

Sometimes, when recipient & donar bloods are mismatched, immediate hemolysis of RBCs occur in the circulating blood.

In this case, antibodies cause the **lysis** of RBCs by activating the complement system & forming a **MAC(membrane attach complex or cytolytic complex)** that inserts itself into the lipid bilayer of cell membranes creating membrane pores permeable to ions & cause osmotic lysis of the cells.

Immediate intravascular hemolysis is far less common than ***agglutination*** followed by ***delayed hemolysis***.

BLOOD TYPING

Before giving a transfusion to a person, it is necessary to determine the blood type of the recipient & donor blood so that the bloods can be appropriately matched. This process is called *blood typing* & *blood matching*.

REQUISITES FOR BLOOD TYPING

To determine the blood group of a person, a suspension of his/her RBCs & testing antisera are required.

Suspension of RBC is prepared by mixing blood drops with isotonic saline(0.9%).

Test sera are:

- 1) Antiserum A containing anti-A antibody.
- 2) Antiserum B containing anti-B antibody.

PROCEDURE OF BLOOD TYPING

- One drop of antiserum A is placed on one end of a glass slide & one drop of antiserum B on the other side.
- One drop of RBC suspension is mixed with each anti-serum. The slide is slightly rocked for two minutes. The presence or absence of agglutination is observed by naked eyes & if necessary it is confirmed by using a microscope.
- Presence of agglutination is confirmed by the presence of thick masses(clumps) of RBCs.
- Absence of agglutination is confirmed by clear mixture with dispersed RBCs.

RESULTS OF BLOOD TYPING

- *If agglutination occurs with antiserum A:* The agglutination occurs if the RBC contains A antigen. So, the blood group is **A**.
- *If agglutination occurs with antiserum B:* The agglutination occurs if the RBC contains B antigen. So, the blood group is **B**.
- *If agglutination occurs with both antisera A & B:* The RBC contain both A & B antigens to cause agglutination. So, the blood group is **AB**.
- *If agglutination does not occur either with antiserum A or antiserum B:* The agglutination does not occur because RBC does not contain any antigen. So, the blood group is **O**.

Rh FACTOR(RHESUS FACTOR)

Rh factor is an antigen present in RBC. This antigen was discovered by *Landsteiner* & *Weiner*. It was first discovered in *Rhesus monkey* & hence named as ***Rh factor***.

There are many Rh antigens but only the ***D antigen*** is more antigenic in human. The persons having D antigen are called ***Rh positive*** & those without D antigen are called ***Rh negative***.

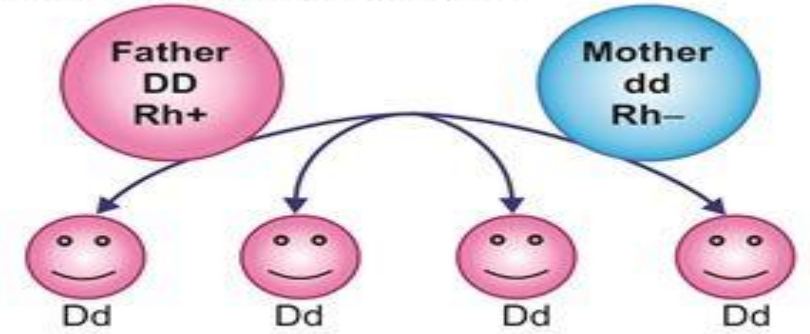
Among Indian population, about 85% of people are Rh+ & 15% are Rh-. Percentage of Rh+ people is more among black people. Worldwide frequencies of Rh+ and Rh- blood types are 95% & 6% respectively.

Rh group system is different from ABO group system because the antigen does not have corresponding natural antibody (anti-D). However, if Rh+ blood is transfused to a Rh- person anti-D is developed in that person. On the other hand, there is no risk of complications if the Rh+ person receives Rh- blood.

INHERITANCE OF Rh ANTIGEN

Rhesus(Rh) factor is an inherited dominant factor. It may be homozygous Rhesus positive with DD or heterozygous Rhesus positive with Dd. Rhesus negative occurs only with complete absence of D(i.e homozygous dd).

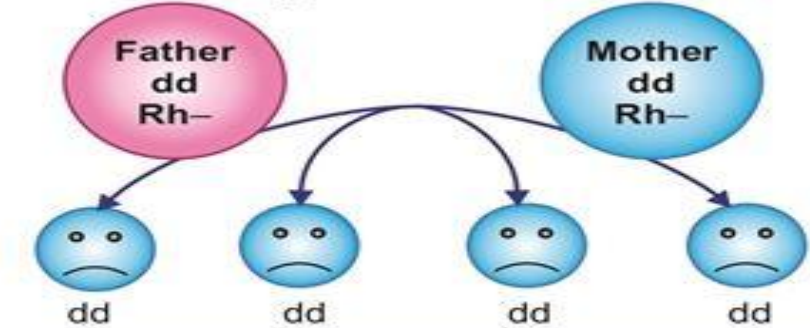
Father – homozygous with DD



A

All offsprings are Rh+ (heterozygous)

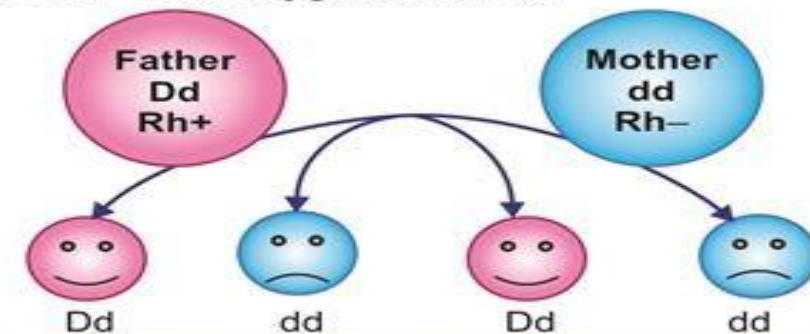
Father – homozygous with dd



B

All offsprings are Rh-

Father – heterozygous with Dd

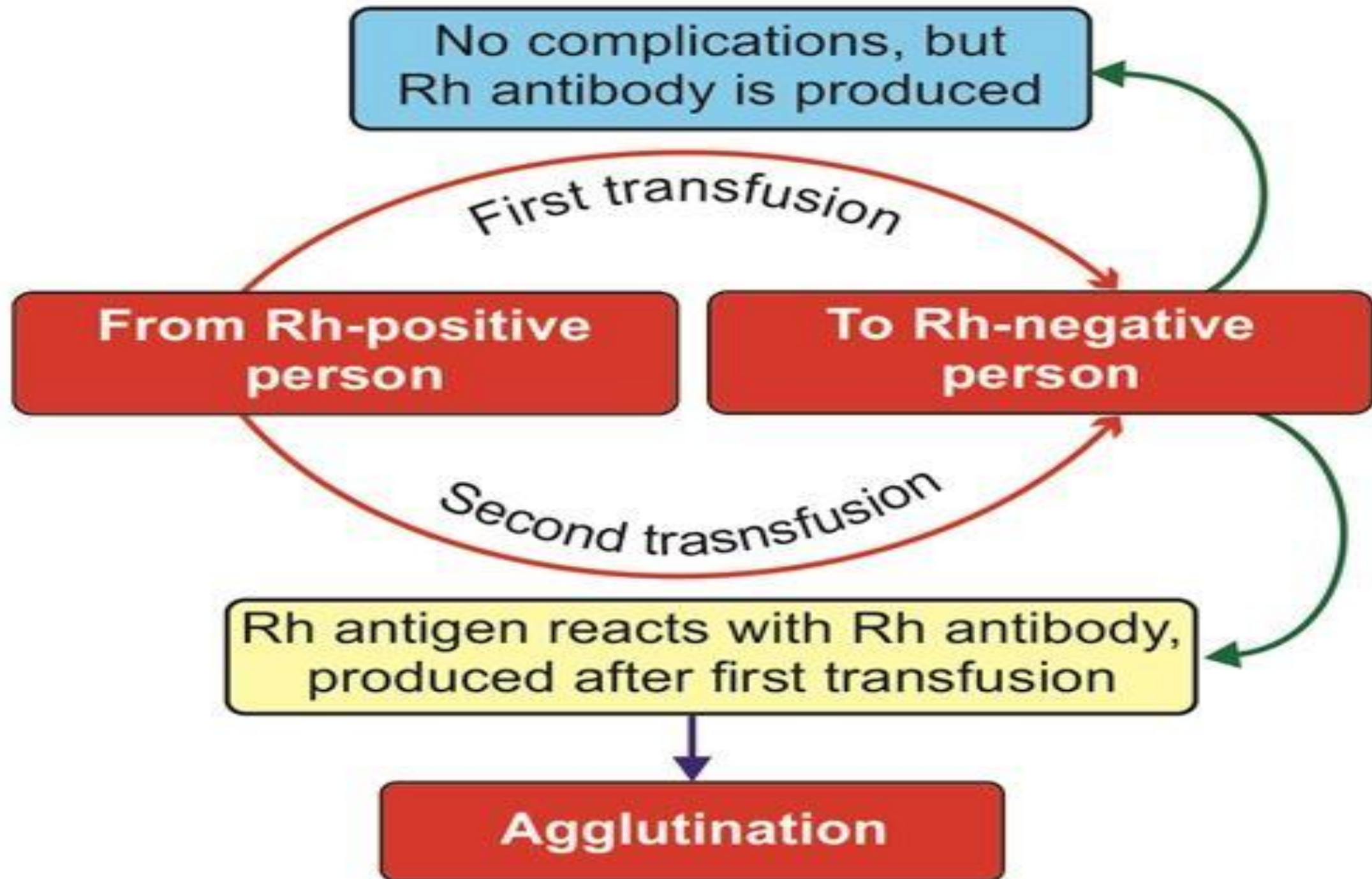


C

50% offsprings are Rh+ (heterozygous)

TRANSFUSION REACTIONS DUE TO Rh INCOMPATIBILITY

When a Rh negative person receives Rh positive blood for the first time, transfusion of Rh positive blood into that person will likely cause no immediate reaction. However, anti-Rh agglutinins develop in sufficient quantities during the next 2 to 4 weeks reaching a maximum concentration of agglutinins about 2 to 4 months later. The transfused RBCs which are still present in the recipient's blood are agglutinated. These agglutinated cells are then hemolyzed by the tissue macrophage system. So, a delayed transfusion reaction occurs. This transfusion reaction is usually mild & does not affect the recipient. However, antibodies developed in the recipient remain in the body forever. On subsequent transfusion of Rh positive blood into the same person who is now already immunized against the Rh factor, the transfusion reaction is greatly enhanced & can be immediate & as severe as a transfusion reaction caused by ABO compatibility.



HEMOLYTIC DISEASE OF FETUS(ERYTHROBLASTOSIS FETALIS)

Hemolytic disease is the disease in fetus & newborn characterized by abnormal hemolysis of RBCs. It is due to Rh incompatibility i.e. the difference between the Rh blood group of the mother & baby.

Hemolytic disease leads to *erythroblastosis fetalis*.

Erythroblastosis fetalis is a disorder in fetus characterized by the presence of erythroblasts in blood. When a mother is Rh negative & the fetus is Rh positive (the Rh factor being inherited from the father), the first child usually escapes the complications of Rh incompatibility because Rh antigen cannot pass from fetal blood into the mother's blood through placental barrier. However, at the time of parturition (delivery of child) the Rh antigen from fetal blood may leak into mother's blood due to placental detachment.

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During the post-partum period i.e. within a month after delivery, the mother develops Rh antibody in her blood. When the mother conceives for the second time & if the fetus going to be Rh positive again, the Rh antibody from mother's blood crosses placental barrier & enters the fetal blood. Rh antibody which enters the fetus causes agglutination of fetal RBCs resulting in hemolysis.

INCIDENCE OF ERYTHROBLASTOSIS FETALIS

An Rh negative mother having her first Rh positive child usually does not develop sufficient anti-Rh agglutinins to cause any harm. However, about 3% of second Rh positive babies exhibit some signs of erythroblastosis fetalis & about 10% of third babies exhibit this disease. The incidence rises progressively with subsequent pregnancies.

CLINICAL PRESENTATION OF ERYTHROBLASTOSIS FETALIS

The jaundiced erythroblastotic newborn baby is usually anaemic at birth & the anti-Rh agglutinins from mother usually circulate in the infant's blood for another 1 to 2 months after birth, destroying more & more RBCs. The hematopoietic tissues of infant attempt to replace the hemolyzed RBCs. The liver & spleen become greatly enlarged & produce RBCs in the same way they normally do during the middle of gestation. Due to the rapid production of RBCs, many early forms of RBCs are passed from baby's bone marrow into the circulation. It is because of the presence of these immature RBCs, the disease is called *erythroblastosis fetalis*.

CHARACTERISTIC FEATURES OF ERYTHROBLASTOSIS FETALIS

Ultimately, due to excessive hemolysis severe complications develop like:

1. **Severe anemia:** excessive hemolysis results in anemia & the infant dies when anemia becomes severe.
2. **Hydrops fetalis:** serious condition characterized by extreme generalized edema in the tissues & organs(ascites, pleural effusion, scalp edema) due to heart failure. In severe cases it may lead to **intrauterine death of fetus.**
3. **Jaundice(hyperbilirubinemia):** rapid breakdown of red blood cells produce high levels of bilirubin causing yellowing of skin & eyes often appearing within 24 to 36 hours of birth.**Kernicterus:**

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4. **Kernicterus:** It is the mental impairment or brain damage in infants due to the precipitation of bilirubin in the neuronal cells due to severe jaundice causing destruction of these cells.
5. **Hepatosplenomegaly:** the liver & spleen become enlarged as they work excessively to produce new red blood cells.
6. **Pallor:** pale skin, lips & nail beds due to severe anemia.
7. **Tachycardia:** rapid heart rate in fetus.
8. **Respiratory distress:** due to pulmonary edema or anemia.
9. **Hypoglycemia:** low blood sugar level potentially due to hyperplasia of pancreatic islet cells.

POSITIVE DIRECT ANTIGLOBULIN TEST(DAT/COOMBS TEST)

It is a laboratory test which confirms maternal antibodies attached to the infant's red blood cells.



TREATMENT OF NEONATES WITH ERYTHROBLASTOSIS FETALIS

If the baby is born with erythroblastosis fetalis, the treatment is to replace the neonatal blood with Rh negative blood. The Rh negative blood is infused over a period of 1.5 or more hours while the neonate's own Rh positive blood is being removed. It will now take at least 6 months for the infant's new Rh positive blood to replace the transfused Rh negative blood. By this time, all the molecules of Rh antibody derived from the mother get destroyed.

PREVENTION OF ERYTHROBLASTOSIS FETALIS

If mother is found to be Rh negative & fetus is Rh positive, anti-D(antibody against D antigen) should be administered to the mother at 28th & 34th weeks of gestation as prophylactic measure.

If Rh negative mother delivers Rh positive baby, then anti-D should be administered to the mother within 48 hrs of delivery. This develops passive immunity & prevents the formation of Rh antibodies in mother's blood. So the hemolytic disease of newborn does not occur in a subsequent pregnancy.

TRANSFUSION REACTION RESULTING FROM MISMATCHED BLOOD TYPES

If donar blood of one type is transfused into a recipient with another blood type, a ***transfusion reaction*** is likely to occur in which the RBCs of donar blood are agglutinated.

ACUTE KIDNEY FAILURE AFTER TRANSFUSION REACTION

One of the most lethal effects of transfusion reactions is kidney failure which can begin within a few minutes to few hours & continue until the person dies of ***acute renal failure***.

ACUTE RENAL FAILURE AFTER MISMATCHED TRANSFUSION

The kidney shutdown seems to have three causes:

1. The antigen antibody reaction of the transfusion reaction release toxic substances from the hemolyzing blood that cause powerful ***renal vasoconstriction***.
2. Loss of circulating RBCs in the recipient along with the production of toxic substances from the hemolyzed cells & the immune reaction often cause ***circulatory shock***. The arterial blood pressure falls very low & renal blood flow decrease.
3. If the total amount of free hemoglobin released into the circulating blood is greater than the quantity that can bind with ***haptoglobin*** (a plasma protein that binds small amounts of hemoglobin), much of the excess leaks through the glomerular membranes into the

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kidney tubules. The tubular hemoglobin concentration rise so high that the hemoglobin precipitates & blocks many of the kidney tubules.

In this way, renal ***vasoconstriction, circulatory shock & blockage of renal tubules*** together cause ***acute renal shutdown***. If the shutdown is complete & fails to resolve, the patient dies within 7 to 12 days.

POST ASSESSMENT

Q1. Cross matching is done to:

- a) Detect infection
- b) Measure hemoglobin
- c) Check compatibility
- d) Increase blood volume

Q2. A person with blood group A has which antibodies:

- a) anti-A
- b) Anti-B
- c) Both anti-A & anti-B
- d) No antibodies

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Q3. Rh factor refers to:

- a) Plasma protein
- b) RBC enzyme
- c) Antigen on RBC surface
- d) White blood cell type

Q4. Erythroblastosis fetalis occurs when:

- a) Mother is Rh+ & fetus is Rh-
- b) Mother is Rh- & fetus is Rh+
- c) Both are Rh+
- d) Both are Rh-

**THANK
YOU**